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Analysing Computer Science Curricula in Nepal's Secondary Schools: **A Management Perspective on Future Workforce Preparedness**[1]

Er. Ajay Kumar Yadav[2]
Dil Prasad Shrestha, PhD[3]

Abstract
This study comprehensively analysed the management of Computer Science (CS) curricula in Nepal's secondary schools and their impact on student career readiness. Data was collected through questionnaire surveys administered to CS teachers and students in Secondary Schools within Kathmandu Valley. The Statistical Package for Social Sciences (SPSS) and Microsoft Excel were utilised for data analysis.

The study underscored the significance of integrating programming, graphics, skill-based pedagogy, artificial intelligence (AI), and interdisciplinary subjects such as Mathematics, Science, and Literature. Teacher competency development initiatives encompassed guidebooks, training programmes, and assessment tests for effective CS instruction.

Furthermore, the study identified textbooks and theoretical/practical sessions as primary learning sources. Recommendations included curriculum enhancements through the incorporation of game-based learning, AI, Graphics, Robotics, and Block-based Programming. While the CS textbook provides moderate coverage of subject units, greater emphasis and support are required for programming, database, and graphics topics due to their inherent complexity.

Keywords: *Computer Science Curriculum, ICT, Education, Policy Managers.*

Introduction

Computer Science is the study of computational thinking, algorithms, data structures, and programming languages, encompassing their own theories, abstractions, and design principles (Comer et al., 1989; Webb et al., 2017). CS is pervasively utilised across various sectors to develop applications for problem-solving and

optimise computational systems. Computational thinking involves the process of decomposing problems into sub-problems, identifying patterns, and devising suitable algorithms (Wing, 2006). Unplugged programming is a pedagogical approach that facilitates learning computational thinking without the use of a physical computer.
The Government of Nepal has emphasised the importance of information communication technology knowledge within the National Education Policy (MoEST, 2019). CS is currently offered as an elective subject in grade 9 in Nepal (CDC, 2019). The Curriculum Development Center (CDC) has designed the CS curriculum in a combined format for both grades 9 and 10.

The CS curriculum comprises 128 teaching hours, evenly distributed between 64 hours of theoretical and practical classes (CDC, 2019). Recently, the CDC has updated the existing CS curricula for both grades 9 and 10, maintaining the same allocation for theoretical and practical sessions, with a total of 128 hours (CDC, 2024).

Objective

This study aims to analyse the Computer Science Curricula for grades 9 and 10 in Nepali secondary schools and recommend improvements for the new curriculum to ensure it equips students with the digital and technological knowledge and skills essential for future workforce preparedness.

Literature Review

The curriculum serves as the official educational framework, encompassing a syllabus, textbook, practical sessions, and defined learning outcomes for the academic year (Ornstein & Hunkins, 2018). Fisher and Frey (2015) emphasise that textbooks provide organised learning resources tailored to the learners' level, interests and needs. Qualitative dimensions of text complexity encompass word, sentence and paragraph meanings in relation to the reader's prior knowledge. Quantitative dimensions include factors such as sentence and paragraph length within a given topic. Readability scores, such as the Flesch Reading Ease and Flesch-Kincaid Grade Level, are employed to assess the text's readability (Flesch, 1948; Zhou et. al., 2007).

A programme is a sequence of instructions designed to execute a specific task. Learning to programme necessitates prerequisite knowledge of language syntax and semantics (Ericson et al., 2022). While visual block-based programming primarily requires semantic knowledge to grasp logic and concepts, text-based programming demands both syntactic and semantic understanding (Weintrop & Wilensky, 2019). Debugging refers to the process of identifying, isolating and rectifying errors within a programme. Artificial Intelligence (AI) involves the development of self-intelligent computers capable of learning, thinking and making decisions akin to humans. AI systems necessitate extensive datasets to facilitate intelligent system functioning. Carbonaro et al. (2010), in a study involving 50 grade 10 students (26 females, 24 males), observed that game-based learning effectively engages both genders in the learning process. Project-based learning fosters student engagement, discussion and collaboration among team members through both verbal and non-verbal communication to achieve project objectives (Hanna, 2008).

The Constitution of Nepal underscores the significance of education and the advancement of science and technology knowledge (MLJPA, 2016). The National Planning Commission's Sixteenth Plan emphasises the crucial role of integrating Information and Communication Technology within the education system to enhance learning outcomes, accessibility and overall educational quality (NPC, 2024). The School Education Sector Plan aims to expand digital infrastructure across all schools, implement teacher development programmes in ICT, and integrate foundational ICT skills into various subject areas (CEHRD, 2016).

Methodology of the Study

Data and information for this study were collected from CS teachers and CS students through an online survey research design. The survey instrument comprised a set of questionnaires to gather feedback on the grades 9 and 10 CS curricula from both teachers and students. Feedback from grades 9 and 10 CS students focused on understanding the pedagogy and textbook content of the CS curriculum, while feedback from grades 11 and 12 CS students aimed to assess the knowledge gained from the CS curriculum.

The survey instrument was structured into two sections: general information and core questions pertaining to the CS curriculum and CS technology. Questions related to textbook units within the CS teacher survey were designed using a five-point Likert scale (Strongly Agree, Agree, Neutral, Disagree, and Strongly Disagree). The Cronbach alpha value for the teacher's survey questionnaire was 0.971, indicating very high reliability and inter-consistency among the teacher responses.

Textbook unit questions for the student survey were similarly designed using a five-point Likert scale (Very Difficult, Difficult, Neutral, Easy, and Very Easy). The Cronbach alpha value for the student survey questionnaire was 0.936, also signifying very high reliability and inter-consistency among student responses. The survey also included open-ended questions concerning pedagogy, contemporary CS technology, and overall suggestions for curriculum improvement. The survey data were subsequently analysed using IBM SPSS Software and MS Excel.

Analysis of Data and Findings

Summary Profile of the Study Participants

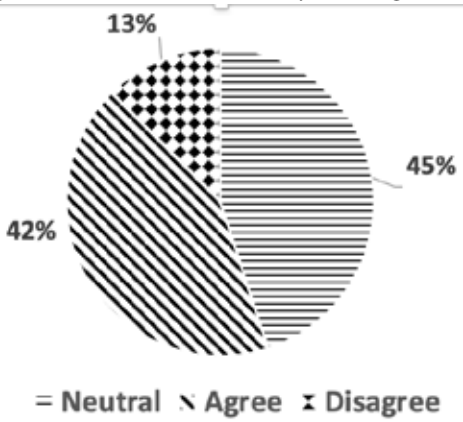
About 419 CS students from 17 schools within Kathmandu Valley participated in an online survey from grades 9 to 12. The study participants comprised boys (61.81%), girls (37.95%), and others (0.24%). The participation rates of students were as follows: Grade 9–(10.26%), Grade 10–(56.32%), Grade 11–(28.89%), and Grade 12–(4.53%).

Similarly, 40 CS teachers from 19 schools participated in an online survey from Kathmandu Valley. The CS teachers' education qualifications consisted of a Bachelor's degree (24%), Master's degree (64%), M.Phil. (8%), and PhD (4%). The CS teachers' specialized subjects were Computer Science (87.5%), Science and Technology (7.5%), Master of Business Studies (2.5%), and Multimedia technologies (2.5%).

Teachers' Perspective on Computer Science Curriculum

CS teachers were asked for their opinions about the CS textbook contents having examples from interdisciplinary subjects such as Mathematics, Statistics, Science, and Literature. Figure 1 presents the teachers' opinions on the integration of interdisciplinary subjects.

Figure 1: CS Curriculum Subject Integrations



Some of the teachers (45%) expressed their balanced views on integrating interdisciplinary subjects in the CS textbook. Similarly, (42%) of the teachers agreed that the CS textbook has reasonably integrated interdisciplinary subjects, and only a very few teachers (13%) indicated that the CS textbook has not integrated interdisciplinary subjects.

CS teachers were requested to provide their views on programming language priorities based on the present CS curricula, and contemporary computer programming languages for grades 9 and 10. Table 1 displays the responses from CS teachers for programming language priorities, based on a multiple-choice question.

Table 1: *Priority Level on Programming Language Priorities.*

Programming Languages	Priority Level (in %)
Python	100
HTML/CSS	93
JavaScript	85
Database SQL	78
QBasic	45

All CS teachers unanimously expressed their view that among the five programming languages, Python should receive the highest priority. The programming language -Python is used for data analysis, data visualization, and artificial intelligence applications. The majority of teachers (93% and 85%) indicated that HTML/CSS and JavaScript should be given the second and third priorities, respectively. HTML and CSS are used for website presentation, whereas JavaScript is used for user interaction in web development. Similarly, Database SQL and QBasic were ranked as the fourth and fifth priorities, respectively.

Teachers were asked about their perspectives on the enhancement of CS curriculum. Table 2 displays their responses (with the multiple-choice question).

Table 2: *Teachers' Perspective on Key Areas for Enhancing the CS Curriculum*

Key Areas	Responses (%)
Programming with a Functional / Application Focus	87
Pedagogy Focus on Skill Development	77
Artificial Intelligence	72
Graphics and Multimedia	70
Alignment with Industry Applications	52
Data Analysis and Visualization	45

A majority of CS teachers (87%) indicated that programming should be taught with a focus on both functional and application-based approaches to build real-world applications. Similarly (77 %, 72%, and 70%) of teachers mentioned that skills-based education, artificial intelligence (AI), graphics and multimedia are required in an enhanced CS curriculum essential for the future workforce preparedness. Some of the teachers (52%) expressed their balanced views for CS curriculum alignment with industry applications that will help to prepare future workforce in the nation-building capacity. Similarly (45%) of the teachers emphasized data analysis and visualization for practical concept implementation of Mathematics and Statistics subject.

CS teachers also pointed out the expected skills of CS students after passing grade 10. Table 3 shows the responses (in a multiple-choice question) provided by the teachers.

Table 3: *Expected Skills of CS Grade 10 Students*

Expected Skills	Responses(%)
CS Concept & Program	85
Knowledge on Computer	82
Digital Literacy	80
Cyber Security	75
Typing skills	70

The significant majority of teachers (85%, 82%, and 80%) responded that the expected skills of CS Grade 10 students include CS Concept & Program, Knowledge of Computers, and Digital Literacy respectively. The CS concepts and programming skills are used to understand algorithms, flowcharts, problem-break, and develop a simple application. The knowledge of the Computer includes understanding different hardware parts of the system, software application of installs/ uninstalls processes, Operating system, file, and folder management of their data. Digital literacy

is all about familiarity with the internet, email, cloud-based applications, and office applications. While (75%) of teachers mentioned the cyber security knowledge related to password safety and severity of data in social media; whereas (70%) of teachers articulated typing skills-(Nepali and English) as expected skills of CS Grade 10 students.

Teachers were also asked for their impressions about some aspects of teacher competency development in connection with the CS subject. Table 4 displays their responses (with the multiple-choice question).

Table 4: *Aspects of Teachers' Competency Development*

Aspects of Teachers' Competency Development	Responses (%)
Teacher Training program	82
Teacher's Guidebook	65
Teachers Assessment Test	47

A majority of the teachers (82%) acknowledged that the teacher training program should receive a high priority. Similarly, some of the teachers (65%) and (47%) mentioned that teacher's guide book and teachers' assessment tests are the crucial aspects of developing teachers' competency required to teach CS subjects, respectively.

Student's Perspective on Computer Science Curriculum

Students were requested to provide their views on the learning experience of CS curriculum with theory and practical classes. Table 5 shows their response to the methods of teaching CS subjects.

Table 5: *Methods of Teaching CS Subject*

Methods of Teaching	Responses (%)
CS theory with comprehensive practical	55
CS theory with some practical	30
CS theory with less practical	15

While 55% of students reported that their CS classes included both theory and adequate practical sessions, others indicated varying levels of practical engagement - 30% mentioned a combination of theory and some practical sessions, and 15% noted theory with limited practical sessions.

Students were surveyed about their understanding of concepts in programming language. Table 6 shows their response to the programming language.

Table 6: *Student Perspectives on Programming Learning*

Programming Learning	Responses (%)
Programming Concepts Focus	40
Basic Coding Skills	24
Application Development Focus	18
Finding It Difficult	18

Around 40% of students have an understanding of programming concepts, 24% of basic coding skills, a few students (18%) expressed confidence in programming and have developed simple applications, and the rest 18% felt programming as a challenging task.

Students were requested to provide their opinions on the CS curriculum enhancement. Table 7 displays their responses based on a multiple-choice question.

Table 7: *Students' Preferences for Topics in an Enhanced Computer Science Curriculum*

Topics	Responses(%)
Game-Based Learning	72
Artificial Intelligence	67
Graphics and Multimedia	64
Robotics	56
Applications Programming	42
Block-Based Program	24

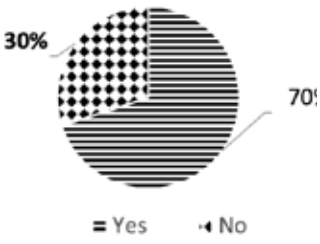
The majority of students (72%, 67%, 64%) mentioned including game-based learning, artificial intelligence, graphics and multimedia (images, audio, and video) in a curriculum

necessary for the future workforce preparedness. Some of the students (56%, and 42%) expressed balanced opinions about including robotics, and application programming in the new curriculum. A few students (24%) indicated to include the block-based program in the new curriculum.

The textbook is one of the primary sources of learning resources at schools. Student's responses to the comprehensiveness of the textbook are presented in Figure 2.

Figure 2: *Textbook Comprehensiveness and Clarity*

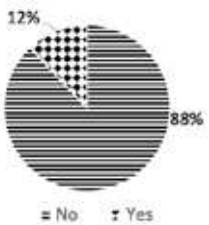
A majority of the students (70%) acknowledged that the textbook is understandable whereas some of them (30%) mentioned that textbook requires including additional examples and URL to explore learning resources.



Students indicated their knowledge and level of Nepali typing that help express their ideas more easily and comprehensively through their native language, especially on topics related to culture and festivals. Figure 3 presents their practices and proficiency in Nepali language typing skills.

Figure 3: *Nepali Language Typing Skills*

A significant majority of the students (88%) lack the skills to type in Nepali language, while a few of students (12%) can type in Nepali language.



CS Teachers' Opinions on the Adequacy of the Textbook Contents

The survey question related to the adequacy of CS curriculum was about to understand whether the textbooks provide sufficient information and examples on topics to cover a wide range of learners. The bar chart illustrates the teachers' opinions on the CS textbook unit's content adequacy for grades 9 and 10. Figures 4 and 5 present the teachers' perspective regarding the adequacy of the textbook contents.

The above bar charts indicate that the textbook contents are largely sufficient for both grades 9 and 10. However, for grade 9, some of the teachers - (25%, 16%, and 16%) suggested the need for some additional examples and solutions in areas such as Programming in QBasic (Unit 4.2), Computer Programming (Unit 4.1), and Computer Graphics (Unit 2.1).

Similarly, for grade 10, a few teachers (18%, 16%, and 16%) recommended expanding contents in Modular Programming (Unit 3.2), File Handling in QBasic (Unit 3.3), and Programming in QBasic (Unit 3.1) with some examples and solutions.

CS Students' Opinions on the Easy and Difficult Units in the Textbook

The survey question related to easy and difficult units was to understand whether the CS textbook has sufficient contents and has been organized from foundational concepts to advanced topics. The bar chart illustrates the students' perceived difficulty levels for various topics in a CS curriculum for grades 9 and 10. Figures 6 and 7 present the students' perspective on easy and difficult units for grades 9 and 10.

The above bar charts show that neutral responses (not too easy and not too difficult) dominate across most unit topics, indicating a balanced perception of difficulty for both grades 9 and 10. For grade 9, Basic Units like Computer Fundamental (Unit 1.1), Types of Computer (Unit 1.2), Basic Architecture of Computer (Unit 1.3), Computer Hardware (Unit 1.4), Computer Software (unit 1.5), and Internet Technology (Unit 3.1) are rated as easier compared to programming and graphics topics like Computer Graphics (Unit

Figure 4: *Teachers' Perspective on the Adequacy of CS Textbook Contents for Grades 9*

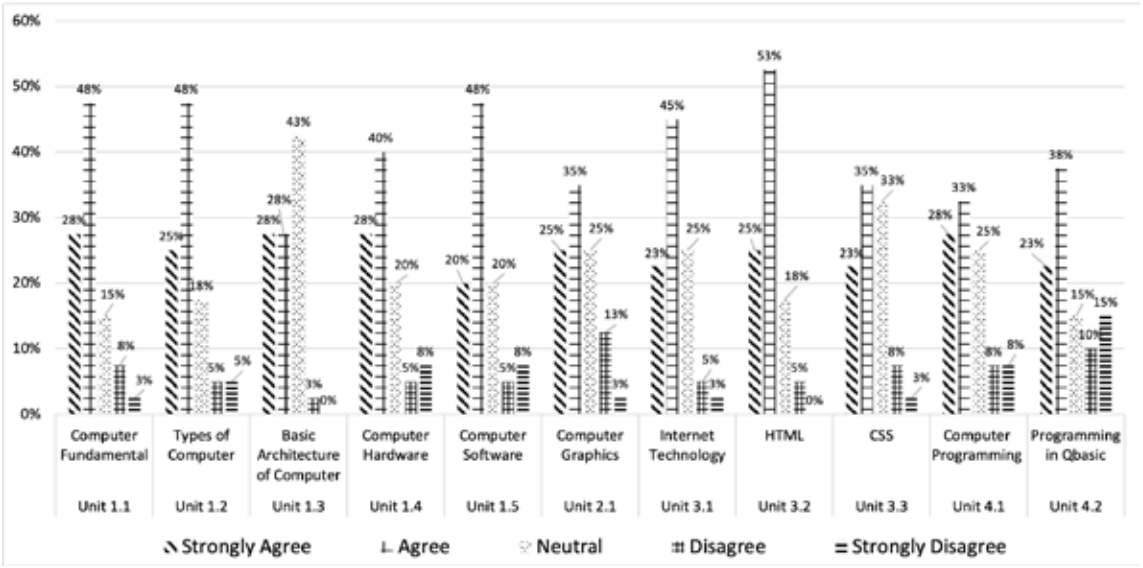


Figure 5: *Teachers' Perspective on the Adequacy of CS Textbook Contents for Grade10*

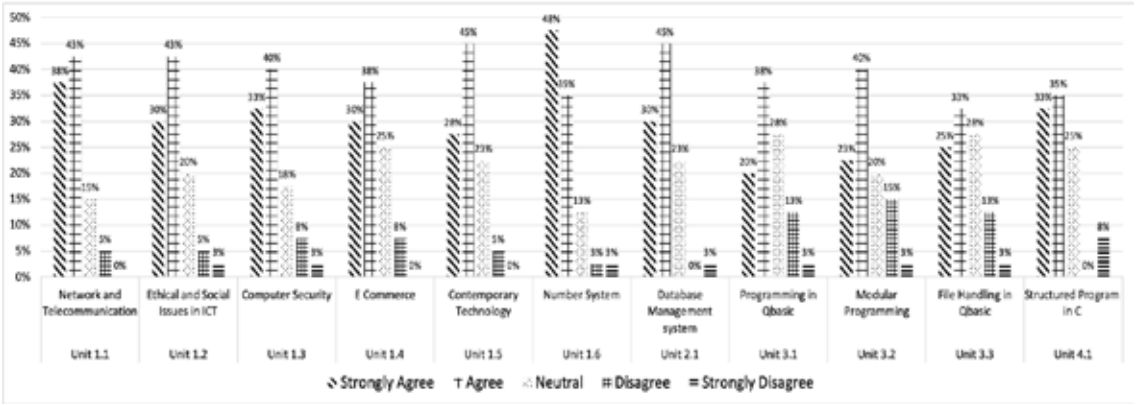


Figure 6: *Students' Perspective on Easy and Difficult Units in the CS Textbook for Grade 9*

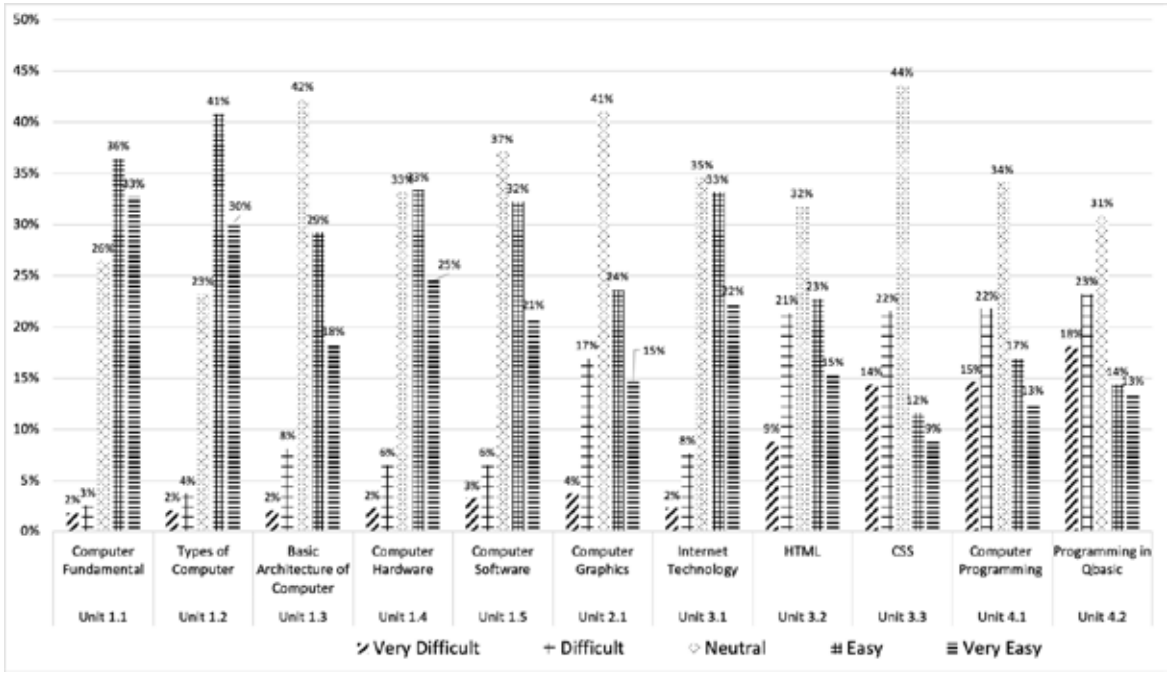
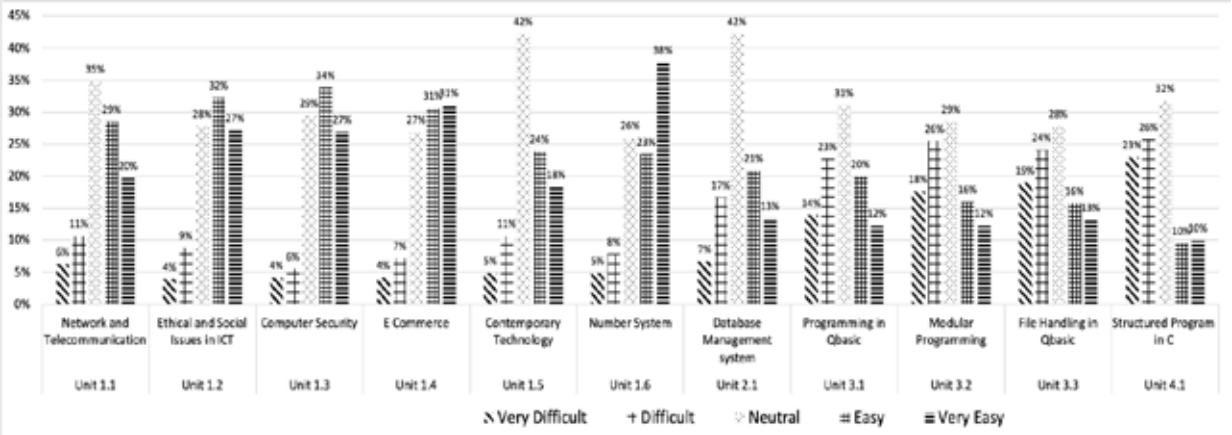


Figure 7: *Students' Perspective on Easy and Difficult Units in the CS Textbook for Grade 10*



2.1), HTML (Unit 3.2), CSS (Unit 3.3), Computer Programming (Unit 4.1), and Programming in QBasic (Unit 4.2) which have higher difficulty ratings.

Similarly for grade 10, Units like Network and Telecommunication (Unit 1.1), Ethical and Social Issues in ICT (Unit 1.2), Computer Security (Unit 1.3), E-Commerce (Unit 1.4), Contemporary Technology (Unit 1.5), and Number System (Unit 1.6) are rated as easier compared to programming and database topics -Database Management System (Unit 2.1), Programming in QBasic (Unit 3.1), Modular Programming (Unit 3.2), File Handling in QBasic (Unit 3.3), and Structured Program in C (Unit 4.1) which have higher difficulty ratings.

The topics related to programming and database seem to require more attention and support, as they show a higher percentage of participants finding them challenging. There is no graphic unit in the grade 10 CS curriculum. The difficult contents are placed after the mid-point of textbooks which usually comes after the mid-term examination in the school.

Results and Discussion

The study found that factors such as programming in practical adaptation, skill-based pedagogy, artificial intelligence, and graphics are important, integrating interdisciplinary subjects such as Mathematics, Statistics, Science, and Literature. Similarly, five programming languages received priority order as follows: Python, HTML/CSS, JavaScript, Database SQL, and the QBasic.

Python is a high-level programming language having several accessible open-source libraries for data analysis, data visualization, and AI applications (Shoaib et al., 2021). The functions can be easily customized to meet specific requirements to develop real-world problem-solving applications. The concepts and examples from interdisciplinary subjects such as Mathematics, Science, Arts, and Literature can be easily implemented through Python programming language, providing several opportunities to learn by doing tasks that are applicable in the real-world context.

JavaScript is also a high-level programming language used for creating user interactions in web development, alongside HTML and CSS. Students grasp CS concepts by implementing user interaction roles in game-based learning.

Skill-based pedagogy emphasizes creating a learning resource aligned with real-world applications. Thus, students are prepared to solve problems with both theoretical knowledge and practical skills, equipping them to address real-world problems.

Expected knowledge and skills of CS grade 10

students include the concept of CS and program, knowledge on computers, digital literacy, cyber security, and typing skills in both Nepali and English.

CS teachers identified a list of CS knowledge and skills essential in the Nepalese social context for students' daily lives and academic careers. The knowledge of Computer hardware, operating systems, and software applications is useful to write, record, organize, solve, and present data in both their job context and academic pursuits.

The study also showed that teacher's guide book, teacher training programs, and teachers' assessment tests are the means of teachers' competency development, which are required to teach the CS subject.

Teachers play a key role in presenting and helping students understand the textbook contents. Therefore, it is essential to provide adequate training to teachers in pedagogy, textbook comprehensive contents, and practical skills. The teacher's guide book includes detailed examples and solutions to various problems, aligned with the local social context. The training program along with assessment tests can critically evaluate learning processes and the effectiveness of the training provided for training of new-generation teachers.

The analyses of data and information provided by the students showed that textbooks, and theoretical and practical sessions emerged as the main sources of learning. They also emphasized the inclusion of Game-based learning, Artificial Intelligence, Graphics, Robotics, Application Programming, and the Block-based Program for the enhancement of the CS Curriculum.

CS concepts can be learned through various teaching-learning methods such as game-based learning, AI, graphics, robotics, application development, and block-based programming. These approaches emphasize different ways to create applications for several purposes to serve diverse students' interests. Since CS is applied across multiple sectors, learning these methods helps students understand and implement CS concepts in real-world scenarios.

The study also concluded that the CS textbook moderately covers subject units, progressing from foundational concepts to advanced topics. However, programming-related topics require greater emphasis and support, as many students find these topics challenging.

Students learn the concept of programming subjectively; requiring diverse methods of learning with practical applications tailored to their preferences and needs. The present CS curriculum has limited learning methods. By incorporating a diverse set of applications, graphics, AI, game-based learning, and robotics; the curriculum may offer various approaches to learning and implementing CS concepts aligning with students' areas of interest.

Conclusion

The CS curriculum has, to a degree, contributed to enhancing digital literacy rates by equipping students with the ICT knowledge and skills essential for the 21st century. Consequently, educational policymakers should consider making CS a mandatory subject rather than offering it as an elective. The current CS curriculum exhibits deficiencies in AI, graphics and real-world application programming knowledge, which are crucial for future workforce and academic success. Furthermore, there is an insufficient number of practical sessions to adequately develop students' problem-solving abilities. A notable shortcoming is the lack of sufficient guidebooks and training programmes for teachers, which hinders teaching effectiveness. In the absence of these resources, teaching efficiency diminishes, leading to suboptimal classroom performance.

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[1] This article is partially based on the thesis entitled "A Study on the Adequacy and Appropriateness of Computer Science Curricula in Nepali Secondary Schools" submitted to Graduate School of Education, Faculty of Education, Tribhuvan University, Kathmandu, Nepal.

[2] Mr. Ajay Kumar Yadav is the Senior Project Manager at a software company and a life member of the Management Association of Nepal (MAN) and the Nepal Engineering Council (NEC). Email – ajayyadav.mit@gmail.com

[3] Dr. Shrestha is a life member of the Management Association of Nepal (MAN) and the Executive Director of Management Innovation, Training, and Research Academy (MITRA) Pvt. Ltd. Email – ilem@ntc.net.np